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ACROSS TIME AND SPACE

VISUALIZING ENVIRONMENTAL CHANGE FROM MULTI-TEMPORAL SATELLITE IMAGES

01 PIPELINE

READING IMAGE

- Sort multi-temporal images by time
- Call multi-image analysis (>2)



PREPROCESSING

- Pairwise comparison with the first image
- Grayscale adjustment



MATCHING

- detectSURFfeatures: feature matching
- estgeform2d: mapping relationship
- imwarp: image warping



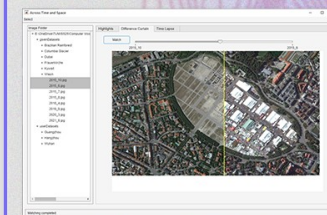
RENDERING

- Align with the first image



02 GUI

THREE VIEWING MODES



- Three viewing modes
- Highlights
- Image Curtain
- Time Lapse



03 PROBLEMS

Fake Matches

Algorithm selection

Image comparison order issues

Global registration failure

Rendering issues

04 METHODS

Adjust parameter thresholds

Use stable and efficient SURF library

Always compare with the first image

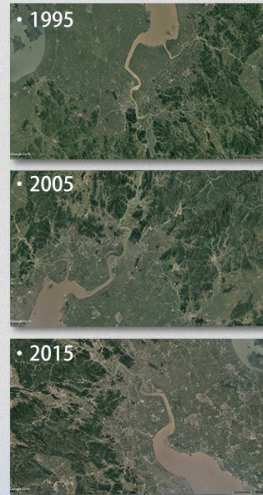
Image subdivision into 4 or 9 parts

Fill black regions

05 RESULTS

Hangzhou, China

ORIGINAL IMAGE



OUTPUT IMAGE



THE 1ST IMAGE

